CS1010: Discrete Mathematics for Computer Science

(Exam-1. Total: 30 marks.)
(Duration: 45 minutes. Date: 09 Sep 2026)

Instructions: ★ You may not get time to answer all the questions unless you have prepared really well. The exam is designed like that. ★ If your mobile phone is found with you during the exam, you will lose one grade. ★ Anybody found copying will get an F grade for the course straight away. ★ It is a no-break exam. You cannot take a break in between. The exam is only for 45 minutes. If you want to go out of the exam hall, you will have to submit your answer paper. ★ You should sit far apart from each other. The halls are big enough. If we see two students sitting close to each other, both the students will lose one grade.

Questions

1. Consider

$$\Phi = (p \to (q \wedge r)) \vee (q \leftrightarrow r).$$

Without using truth tables, answer the following.

(a) Using logical equivalence laws taught in class (you don't need to remember the names of these laws), simplify Φ to $\neg p \vee (q \leftrightarrow r)$. [2 marks]

(b) Obtain both a conjunctive normal form (CNF) and a disjunctive normal form (DNF) for Φ. [2 marks]

2. The domain is a group containing at least two researchers. The relation $C(x, y)$ means "researcher x has collaborated with researcher y." Do not assume that C is symmetric.

(a) Translate the following statements into predicate logic. [2 marks]

i. Every researcher has collaborated with at least one other researcher.

ii. There is a researcher with whom every other researcher has collaborated.

(b) Negate each formula from part (a), with no negation sign occurring outside a quantifier. Also express each negation in English. [2 marks]

3. Let a and b be odd integers.

(a) Prove directly that $a^2 - b^2$ is divisible by 8. [2 marks]

(b) State the converse. Is it true? Supply either a proof or a counterexample. [2 marks]

4. Suppose you built a machine that tells "lucky" numbers from "unlucky" numbers. That is, given a natural number, the machine will respond with either "lucky" or "unlucky". Two such machines are considered different if there is at least one natural number for which one machine responds "lucky" while the other responds "unlucky".

   Use Cantor's diagonal argument to prove that there are uncountably many distinct machines. [2 marks]

5. Let N denote the set of all natural numbers. Let S denote the set of all finite subsets of N. What is the cardinality of S? Explain your answer. [3 marks]

6. Give an example to show that

   (a) the intersection of two countably infinite sets may also be countably infinite.

   (b) the intersection of two countably infinite sets may be finite. [2 marks]

7. Let $\varphi_2(n)$ denote the number of positive integers a such that both a and $a+1$ are relatively prime to n. Find a formula for $\varphi_2(n)$ where n is odd. You may assume that $\varphi_2(n)$ is a multiplicative function; that is, if $\gcd(m,n)=1$, then

$$\varphi_2(mn) = \varphi_2(m)\varphi_2(n).$$

[3 marks]

8. **TRUE or FALSE:** *(One word answer. No need to give any explanation.)*
   Assume a, b, c are positive integers unless otherwise stated.

   (a) If $\gcd(a,b)=1$ and $\gcd(a,c)=1$, then $\gcd(a,bc)=1$. T

   (b) If $\gcd(a,b)=1$, then $\gcd(a+b,a-b)=1$. F

   (c) If $a \mid b$ and $\gcd(a,c)=1$, then $\gcd(b,c)=1$. F

   (d) If $a \equiv b \pmod{n}$, then $a^k \equiv b^k \pmod{n}$ for every positive integer k. T

   (e) If $a^2 \equiv b^2 \pmod{n}$, then $a \equiv b \pmod{n}$. F

   (f) If $a^2 \equiv 1 \pmod{p}$, where p is prime, then $a \equiv 1 \pmod{p}$. F

   (g) If $a \mid b$ and $a \mid c$, then $a \mid (b+c)$. T

   (h) $\gcd(a,b) = \gcd(b, a \bmod b)$. T

[8 mark]

ALL THE BEST